1. Which ETF has the highest annualized rate of return?

   A. ETF 1

   B. ETF 2

   C. ETF 3

   Solution:

   B is correct. The annualized rate of return for the three ETFs are as follows:

   ETF 1 annualized return = $(1.0461^{365/146}) - 1 = 11.93\%$

   ETF 2 annualized return = $(1.0110^{52/5}) - 1 = 12.05\%$

   ETF 3 annualized return = $(1.1435^{12/15}) - 1 = 11.32\%$

   Despite having the lowest value for the periodic rate, ETF 2 has the highest annualized rate of return because of the reinvestment rate assumption and the compounding of the periodic rate.

Continuously Compounded Returns

An important concept is the continuously compounded return associated with a holding period return, such as R_1. The **continuously compounded return** associated with a holding period return is the natural logarithm of one plus that holding period return, or equivalently, the natural logarithm of the ending price over the beginning price (the price relative). Note that here we are using r to refer specifically to continuously compounded returns, but other textbooks and sources may use a different notation.

If we observe a one-week holding period return of 0.04, the equivalent continuously compounded return, called the one-week continuously compounded return, is ln(1.04) = 0.039221; EUR1.00 invested for one week at 0.039221 continuously compounded gives EUR1.04, equivalent to a 4 percent one-week holding period return.

The continuously compounded return from t to $t + 1$ is

$$r_{t,t+1} = \ln(P_{t+1}/P_t) = \ln(1 + R_{t,t+1}). \tag{10}$$

For our example, an asset purchased at time t for a P_0 of USD30 and the same asset one period later, $t + 1$, has a value of P_1 of USD34.50 has a continuously compounded return given by $r_{0,1} = \ln(P_1/P_0) = \ln(1 + R_{0,1}) = \ln(\text{USD}34.50/\text{USD}30) = \ln(1.15) = 0.139762$.

Thus, 13.98 percent is the continuously compounded return from $t = 0$ to $t = 1$. The continuously compounded return is smaller than the associated holding period return. If our investment horizon extends from $t = 0$ to $t = T$, then the continuously compounded return to T is

$$r_{0,\mathrm{T}} = \ln(P_T/P_0). \tag{11}$$

Applying the exponential function to both sides of the equation, we have $\exp(r_{0,\mathrm{T}}) = \exp[\ln(P_T/P_0)] = P_T/P_0$, so

$$P_T = P_0\exp(r_{0,\mathrm{T}}).$$

We can also express P_T/P_0 as the product of price relatives:

$$P_T/P_0 = (P_T/P_{T-1})(P_{T-1}/P_{T-2}) \ldots (P_1/P_0). \tag{12}$$

Taking logs of both sides of this equation, we find that the continuously compounded return to time T is the sum of the one-period continuously compounded returns:

$$r_{0,\mathrm{T}} = r_{T-1,T} + r_{T-2,T-1} + \ldots + r_{0,1}. \tag{13}$$